EE-482: Network Security and Cryptography

LECTURE NO 10: *RSA*

Engr. Tehseen Ahsan
Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan

Organization of Lecture

1. Introduction
2. RSA algorithm

1. Introduction (1/2)

→ Invented by Rivest Shamir Adleman in 1977

→ Most popular public key cryptosystem

→ Based on prime number.

A prime number is one that is divisible by 1 and itself.

2, 3, 5, 7, 11, 13, 17 ⟶ Prime.nos

4, 6, 8, 10, 12 etc ⟶ Not prime nos

A prime no. above 2 must be an ODD no.

1. Introduction (2/2)

Mathematical Fact: it is easy to find and multiply large prime nos. but extremely difficult to factor their product

→ The private and public keys in RSA are based on very large (100 digits) prime numbers.

→ RSA algorithm is simple, however the challenge lies in selection & generation of public & private keys.

→ Hardware RSA is 1000 times slower than hardware DES.

→ Software RSA is 100 times slower than software DES.

→ RSA is very difficult to break if key is secret.

2. RSA algorithm (1/20)

RSA Algorithm

Step 1: Choose two large prime numbers P & Q and keep them secret.

Step 2: Calculate: $N = P \times Q \longrightarrow$ Public

Step 3: Select public key (encryption key E) such that it is not a factor of $(P-1)(Q-1)$

Step 4: Select private key (decryption key D) such that

$$(D \times E) \bmod (P-1)(Q-1) = 1$$

Step 5: For encryption calculate: $C.T = P.T^{E} \bmod N$

Step 6: For decryption calculate: $P.T = C.T^{D} \bmod N$

2. RSA algorithm (2/20)

↳ E & D are different ⟹ asymmetric

↳ E & D are related to each other

↳ E: Public key & known

D: Private key & kept secret

Secrecy of the algorithm lies in 'D'.

2. RSA algorithm (3/20)

Example 1

1) Prime nos: $P = 7$, $Q = 17 \Longrightarrow$ secret

2) $N = P \times Q = 7 \times 17 =$ **119** $\Longrightarrow$ Public

3) $(P-1)(Q-1) = (7-1)(17-1) =$ **96**

2	96
2	48-0
2	24-0
2	12-0
2	6-0
3	3-0
	1-0

Factors of 96 are: $2 \times 2 \times 2 \times 2 \times 2 \times 3$

Public key 'E' must not have a factor of 2 & 3

i.e 'E' cannot be 2, 4, 6, 9, 10, ...

2×1 2×2 2×3 3×3 2×5 or 5×2

Choose $E = 5$

2. RSA algorithm (4/20)

4) Private key 'D' must satisfy

$(D \times E) \bmod (P-1)(Q-1) = 1$

$D \times 5 \bmod 96 = 1$

Say $D = 77$

$77 \times 5 \bmod 96 = 1$

$385 \bmod 96 = 1$

$$\begin{array}{r|l} & 4 \rightarrow q \\ 96 & 385 \\ & 384 \\ \hline & 1 \rightarrow r \end{array}$$

Note: Must choose 'D' to obtain remainder 1

Public key, $E = 5 \longrightarrow$ published

Private key, $D = 77 \longrightarrow$ kept secret

Say Alice wants to send P.T = 6 to Bob

2. RSA algorithm (5/20)

→ Alice will encrypt P.T using Bob's public key E:

$$C.T = P.T^{E} \bmod N$$

$$= 6^{5} \bmod 119$$

$$= 7776 \bmod 119$$

$$\boxed{C.T = 41 \text{ (Encrypted \& sent)}}$$

→ At receiver Bob will decrypt using secret key D.

$$P.T = C.T^{D} \bmod N$$

$$= 41^{77} \bmod 119 \rightarrow$$

Solve it using Fast exponential method as discussed in Diffie-Helman Algorithm previously

$$\boxed{P.T = 6}$$

2. RSA algorithm (6/20)

Extension of Example 1:

* Suppose we have been asked to encrypt & decrypt the given message using private / public key pair i.e;

↳ Message : P I A ⇒ $M_1 \; M_2 \; M_3$

↳ Public key, $E = 3$

↳ Private key, $D = 27$

→ $M_1 = P$; $M_2 = I$; $M_3 = A$

1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L
13	14	15	16	17	18	19	20	21	22	23	24
M	N	O	P	Q	R	S	T	U	V	W	X
25	26										
Y	Z										

2. RSA algorithm (7/20)

Encryption:

↳ $M_1 = P = 16$

→ Cipher Text for Message M_1 will be

→ $C.T_1 = M_1^E \bmod N$

or

$C.T_1 = P.T_1^E \bmod N$

$= 16^3 \bmod 55 = 4096 \bmod 55 = 26$

$C.T_1 = 26 = Z$

$\boxed{C.T_1 = Z}$

2. RSA algorithm (8/20)

Similarly

$\rightarrow C.T_2 = M_2^E \bmod N \quad ; \; M_2 = I = 9$

$= 9^3 \bmod 55 = 729 \bmod 55$

$= 14 = N$

$C.T_2 = 14 = N$

$\boxed{C.T_2 = N}$

$\rightarrow C.T_3 = M_3^E \bmod N \quad ; \; M_3 = A = 1$

$= A^E \bmod N$

$= 1^3 \bmod 55$

$= 1$

$\boxed{C.T_3 = A}$

2. RSA algorithm (9/20)

Decryption

$\rightarrow M_1 = C.T_1^D \bmod N$

$= Z^D \bmod N$

$M_1 = 26^{27} \bmod 55 \Rightarrow$ (Solve it using Fast Exponential method as discussed in Deffie-Hellman Algorithm)

$\rightarrow M_2 = C.T_2^D \bmod N$

$= N^D \bmod N$

$= 14^{27} \bmod 55 \rightarrow$ (Same as in M_1 case)

$\rightarrow M_3 = C.T_3^D \bmod N$

$= A^D \bmod N$

$= 1^{27} \bmod 55 = 1 \bmod 55 = 1 = A$; $\boxed{M_3 = A}$

2. RSA algorithm (10/20)

Example 2 (H.W)

Given.

$P = 3$, $Q = 11$ & $P.T = 4$

Apply RSA algorithm on the given data

Security of RSA

→ Attacker has knowledge of E & N.
→ in order to break need 'D'
→ To find 'D' need knowledge of P & Q
→ P & Q are difficult to find if 'N' is very Large
→ cannot find 'D' knowing E & N.

2. RSA algorithm (11/20)

How to find "D" for step 4 of RSA Algorithm?

Use Euclid's Algorithm

↳ Euclid's Algorithm finds the greatest common divisor (GCD) of two numbers.

↳ GCD: largest integer that evenly divides both of them

2. RSA algorithm (12/20)

Example3 find GCD of 408 & 595 using Euclid's Algorithm.

step quotient remainder

n	q_n	r_n	U_n	
Initial Steps -2		E → 408	1	
-1	(P-1)(Q-1) → 595		0	
0	$q_0 = 0$	408	1	
1	$q_1 = 1$	187	-1	
2	$q_2 = 2$	34	3	
3	$q_3 = 5$	$\boxed{17}$ gcd	-16	
4	$q_4 = 2$	0		

* U_n: initialize by 1 & 0 i.e $u_{-2} = 1$ & $u_{-1} = 0$

* At step n: $U_n = U_{n-2} - q_n u_{n-1}$

2. RSA algorithm (13/20)

at $n=0$:

$$u_0 = u_{-2} - q_0 u_{-1} = 1 - 0 = 1$$

$n=1$:

$$u_1 = u_{-1} - q_1 u_0 = 0 - 1(1) = -1$$

$n=2$:

$$u_2 = u_0 - q_2 u_1 = 1 - 2(-1) = 1+2 = 3$$

$n=3$:

$$u_3 = u_1 - q_3 u_2 = -1 - (5)(3) = -1 \quad 15 = -16$$

Therefore decryption key "D" can be computed as

$$D = (P-1)(Q-1) - u_3$$

$$= 595 - 16$$

$$\boxed{D = 579}$$

in this example $(P-1)(Q-1) = 595$

2. RSA algorithm (14/20)

Recall step 4 of RSA which states:

$$(D \times E) \bmod (P-1)(Q-1) = 1 \longrightarrow (1)$$

In above equation (1) we have

$E = 408$

$D = 579$

$(P-1)(Q-1) = 595$

$\gcd(408, 595) = 17$

$$(1) \Rightarrow (579 \times 408) \bmod 595 = 1$$

$$236232 \bmod 595 = 1$$

Hence for value of $D = 579$, eq 1 is verified

2. RSA algorithm (15/20)

<u>Revisiting Example 1</u>:

1) $P = 7$, $Q = 17$

2) $N = P \times Q = 7 \times 17 = 119$

3) $(P-1)(Q-1) = 6 \times 16 = 96$

Choose $E = 5$

4) $D = ?$

2. RSA algorithm (16/20)

Soln

To find D:

$$D \times E \bmod 96 = 1$$

$$D \times 5 \bmod 96 = 1 \rightarrow (A)$$

Using Euclid Algorithm

E

(P-1)(Q-1)

n	q_n	r_n	u_n
-2	—	5	1
-1	—	96	0
0	0	5	1
1	19	$\boxed{1}$ gcd	-19
2	5	0	

$$U_n = U_{n-2} - q_n U_{n-1}$$

$$U_0 = U_{-2} - q_0 U_{-1} = 1 - 0 = 1$$

$$U_1 = U_{-1} - q_1 U_0 = 0 - 19(1) = -19$$

So $D = (P-1)(Q-1) - U_1 = 96 - 19 = 77$

2. RSA algorithm (17/20)

Eq Ⓐ $\Rightarrow D \times 5 \bmod 96 = 1$

$77 \times 5 \bmod 96 = 1$

$\boxed{385 \bmod 96 = 1}$ verified

NOTE : • if u_n is negative subtract from $(P-1)(Q-1)$

• if u_n is positive remain as it is

2. RSA algorithm (18/20)

Example 4:

2	192
2	96 – 0
2	48 – 0
2	24 – 0
2	12 – 0
2	6 – 0
3	3 – 0
	1 – 0

1) $P = 13$ & $Q = 17$

2) $N = P \times Q = 13 \times 17 = 221$

3) $(P-1)(Q-1) = (13-1)(17-1) = 192$

4) choose "E" such that it is not a factor of $(P-1)(Q-1)$ i.e "E" must not have a factor of 2 & 3; So choose $E = 5$

5) Decryption key $D = ?$

choose "D" such that:

$D \times E \bmod 192 = 1$

$D \times 5 \bmod 192 = 1 \longrightarrow$ (A)

2. RSA algorithm (19/20)

n	q_n	r_n	v_n
−2	−	5	1
−1	−	192	0
0	0	5	1
1	38	2	−38
2	2	$\boxed{1}$ gcd	77
3	2	0	

$v_0 = v_{-2} - q_0 v_{-1} = 1 - 0 = \boxed{1}$

$v_1 = v_{-1} - q_1 v_0 = 0 - 38(1) = \boxed{-38}$

$v_2 = v_0 - q_2 v_1 = 1 - 2(-38) = 77$

As v_n (v_2) is +ve we will choose "D" as

$\therefore D = v_2 = 77$; $\boxed{D = 77}$

Eq ④ ⇒ $D \times 5 \bmod 192 = 1$

$77 \times 5 \bmod 192 = 1$

$\boxed{385 \bmod 192 = 1}$ verified

2. RSA algorithm (20/20)

Example 5 (H.W)

P = 3, Q = 11

* Choose an appropiate value for encryption key "E" using RSA algorithm
* calculate value of decryption key "D" using Euclid Algorithm in RSA.

References: Textbooks

1. *"Cryptography and Network Security", by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *"Cryptography and Network Security Principles and Practice", by William stallings, 6th Edition, Pearson.*
3. *"Computer Security Principles and Practice", by William stallings, Lawrie Brown, 4th Edition, Pearson.*